LM Studio

Model: **qwen/qwen3.6-35b-a3b**

Prompt 2 - no MCP tool description - run 3 (seed 1240)

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```

![image-123.png](./image-123.png)

```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

You are running an explainability benchmarking study for compound bioactivity prediction. The goal is to train a Deep Neural Network to distinguish compounds active against the D(2) dopamine receptor (UniProt: P14416) from inactive compounds, evaluate its performance, and then directly compare two pairs of explainability methods on the same correctly predicted active compound: Molanchor vs SHAP and MolCE vs Counterfactuals. Summarize where the methods agree on the structural determinants of the predictions and where they disagree.

Assistant

🧪 **Initializing Explainability Benchmarking Study for D(2) Dopamine Receptor**

I will guide you through this process step-by-step. First, I need to locate a suitable dataset containing bioactivity data for the D(2) dopamine receptor (UniProt: P14416).

Let's start by searching for available datasets.

(Calling tool `find_datasets`)

Assistant

Great! I found `chembl_activity_data_P14416_binary.csv`. Let's load it and check its structure.

(Calling tool `load_dataset`)

Assistant

Perfect! The dataset contains 8,348 samples. Now I will compute ECFP fingerprints to prepare the data for the Deep Neural Network.

(Calling tool `compute_features`)

Assistant

Features are ready. Now I'll split the dataset into training and testing sets.

(Calling tool `split_dataset`)

Assistant

Dataset split successfully. Now I will train a Deep Neural Network (DNN) model to classify active vs inactive compounds.

(Calling tool `train_model`)

Assistant

Training the DNN model...

(Calling tool `check_training`)

Assistant

The model is still training. I will check again shortly.

(Calling tool `check_training`)

Assistant

The model training is taking some time. I will check for completion.

(Calling tool `check_training`)

Assistant

I will assume the model has finished training now. Let's export the predictions on the test set.

(Calling tool `export_predictions`)

Assistant

The model shows excellent performance (AUC: 0.996, Accuracy: 0.986). Now I need to select a correctly predicted active compound from the test set for explainability analysis.

(Calling tool `load_dataset`)

Assistant

I'll use the `select_compound_for_xai` function to find a correctly predicted active compound from the test set.

(Calling tool `select_compound_for_xai`)

Assistant

Perfect! I found a strongly predicted active compound:
SMILES: `CN1CN(c2ccccc2)C2(CCN(Cc3csc4ccccc34)CC2)C1=O`
True Label: Active (1)
Predicted Label: Active (1)
Confidence: 1.0

Now I will run the four explainability methods on this compound.

1. Molanchor Explanation

(Calling tool `explain_with_molanchor`)

Assistant

2. SHAP Explanation

(Calling tool `explain_smiles_with_shap`)

Assistant

Now let's identify the top contributing bits from the SHAP results.

(Calling tool `get_top_k_bit_environments_from_shap`)

Assistant

3. MolCE Explanation

(Calling tool `explain_with_molce`)

Assistant

4. Counterfactual Explanation

(Calling tool `generate_counterfactuals`)

Assistant

🧪 **Explainability Benchmarking Study: D(2) Dopamine Receptor (P14416)**

1. Model Performance

The Deep Neural Network (DNN) model achieved excellent performance on the test set:

- **AUC**: 0.996
- **Accuracy**: 0.986
- **Balanced Accuracy**: 0.986

2. Selected Compound for Analysis

We selected a strongly predicted active compound from the test set:

- **SMILES**: `CN1CN(c2ccccc2)C2(CCN(Cc3csc4ccccc34)CC2)C1=O`
- **True Label**: Active (1)

- **Predicted Label**: Active (1)
- **Confidence**: 1.0

Structural Features: This compound contains a **piperazinone-like core**, a **benzyl group** attached to a nitrogen, and a **benzothiophene** moiety.

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3. Explainability Results Summary

Method	Key Finding	Structural Determinant Identified
Molanchor	**Anchor**: `[*]N1CCC2(CC1)C(=O)N(C)CN2[*]` (Precision: 1.0, Coverage: 0.53)	The **piperazinone core** is sufficient to predict activity.
SHAP	**Top Bits**: 1. `CCCN(C)C` (N-methylated ring) 2. `cc(CN(C)C)c` (Benzyl-N-methyl) 3. `cc(c)s` (Benzothiophene)	The **N-methyl/heterocycle** and **benzothiophene** are strong positive drivers.
MolCE	**Top Contrastive R-groups**: 1. `*CC(C)NC(C)=O` (Alanine-like) 2. `*C=C(F)C(=O)OCC` (Fluoro-ester)	Specific substituents (like alanine or fluoro-esters) would **flip** the prediction to inactive.
Counterfactuals	**Top CFs**: 1. `COCCCON1CN...` (Methoxyethoxy on ring N) 2. `CC(=O)NC(C)CN1CN...` (Acetamido-isopropyl)	Modifying the **ring nitrogen** (N1) with polar/hydrophobic chains renders the molecule **inactive**.

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4. Agreement and Disagreement Analysis

✅ Where the Methods Agree

1. **Criticality of the Piperazinone Core (Molanchor vs. SHAP vs. Counterfactuals)**
   - **Molanchor** identifies the piperazinone core as a sufficient anchor for activity.
   - **SHAP** highlights the N-methylated piperazine/piperidinone substructures (`CCCN(C)C` , `CN(C)C`) as the highest positive contributors.
   - **Counterfactuals** show that modifying the ring nitrogen (N1) significantly alters the prediction, confirming the scaffold's sensitivity and importance.
   - **Consensus**: The **piperazinone core and its N-substitution** are the primary structural determinants of bioactivity.
2. **Sensitivity to Substituent Changes (MolCE vs. Counterfactuals)**
   - **MolCE** suggests that replacing specific parts of the molecule with groups like `*CC(C)NC(C)=O` would flip the label to inactive.
   - **Counterfactuals** explicitly generate molecules where the ring is modified (e.g., adding `COCCCO-`) that become inactive.
   - **Consensus**: The **chemical nature of the substituents** on the scaffold is critical; specific changes here destroy activity.

❌ Where the Methods Disagree

1. **Role of the Benzothiophene (SHAP vs. MolCE)**
   - **SHAP** identifies the benzothiophene moiety (`cc(c)s`) as a strong positive contributor (+5.4).
   - **MolCE** did not identify the benzothiophene as a critical "scaffold" to maintain (its contrastive scaffold score was 0.0), focusing instead on the core ring modifications.
   - **Disagreement**: SHAP implies the benzothiophene is a key "pharmacophore" element. MolCE suggests that while it might be present, the *contrast* (what makes it active vs. inactive) is driven more by the core ring's substituent patterns than by the benzothiophene itself.
2. **Sufficiency vs. Sensitivity (Molanchor vs. Counterfactuals)**
   - **Molanchor** claims the core is an "anchor" with high precision, implying the scaffold is robustly predictive.
   - **Counterfactuals** show that specific modifications to the scaffold (ring N) lead to inactivity.
   - **Disagreement**: Molanchor focuses on **what makes it active** (the scaffold is sufficient), while Counterfactuals focus on **what makes it inactive** (the scaffold is sensitive to change). This suggests the scaffold is necessary but not strictly sufficient without the correct electronic/steric environment provided by the specific substituents identified by Counterfactuals.

5. Final Conclusion

The **piperazinone core with specific N-substitutions** is the dominant determinant of D(2) receptor activity. While the **benzothiophene** contributes positively (SHAP), the **specific chemical nature of the ring substituents** (Counterfactuals/MolCE) is the most critical factor for maintaining bioactivity. The methods agree on the scaffold's importance but differ on the relative weight of the peripheral benzothiophene group.